Multimodal Emotion Recognition with Adaptive Fusion and Reliability-Aware Attention using Lightweight Transformer Architectures

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Abstract

Multimodal emotion recognition has emerged as a key research area within affective computing, driven by the need to interpret human emotions from heterogeneous sources such as text, speech, and facial expressions. Recent advances in deep learning, particularly transformer-based architectures, have significantly improved performance by enabling rich cross-modal interactions. However, a critical review of existing literature reveals that many state-of-the-art systems rely on computationally intensive models, exhibit sensitivity to noisy or missing modalities, and perform poorly on underrepresented emotion categories. This paper reviews recent developments in multimodal emotion recognition, focusing on fusion strategies, transformer-based models, robustness mechanisms, and approaches for handling class imbalance. By synthesizing findings across multiple research directions, the review highlights key limitations in current systems and identifies emerging trends toward lightweight, reliability-aware, and generalizable fusion frameworks. The analysis provides a consolidated perspective on the state of the art while motivating the design of efficient and robust multimodal emotion recognition systems suitable for real-world deployment.

1. Introduction

1.1 Multimodal Emotion Recognition: Evolving Challenges

The increasing deployment of intelligent systems in domains such as human–computer interaction, mental health analysis, and social media analytics has intensified the need for accurate emotion recognition. Human emotions are rarely conveyed through a single signal; instead, they emerge from the combined interaction of language, vocal patterns, and facial expressions. Although multimodal learning has shown promise in capturing this complexity, existing emotion recognition systems continue to face significant challenges. Models trained under controlled conditions often struggle when deployed in real-world environments where inputs may be noisy, incomplete, or temporally misaligned. Additionally, the growing reliance on deep learning architectures has introduced substantial computational overhead, limiting scalability and real-time applicability [1], [2]. These challenges have motivated researchers to explore more efficient, robust, and adaptable multimodal emotion recognition frameworks.

1.2 Fusion and Generalization Across Modalities

Most contemporary multimodal emotion recognition systems employ separate encoders for text, audio, and visual data, followed by a fusion mechanism that combines modality-specific representations. While this design improves performance over unimodal approaches, the fusion process often remains tightly coupled to a specific dataset or input configuration. As a result, models trained on one distribution may generalize poorly when modalities differ in quality or availability. Transformer-based fusion methods attempt to address this limitation by modeling cross-modal interactions through attention mechanisms, enabling richer contextual alignment across modalities [3]. However, many existing fusion architectures assume the consistent presence of all modalities and prioritize accuracy over adaptability, leading to degraded performance when one or more modalities are unreliable or missing [4]. This highlights a gap in current systems, where effective cross-modal learning must be balanced with robustness and generalization.

1.3 Robustness and Privacy-Inspired Reliability Mechanisms

Recent research trends emphasize the importance of reliability-aware modeling in multimodal systems. Techniques inspired by uncertainty estimation and robustness analysis aim to assess the quality of each modality before fusion, allowing models to down-weight noisy or uninformative inputs [5]. In parallel, data-centric approaches such as targeted augmentation and regularization seek to improve model stability, particularly for underrepresented emotion categories [6]. These methods are especially valuable in practical settings, where data imperfections are unavoidable and emotion distributions are inherently imbalanced. Incorporating reliability-aware attention mechanisms alongside lightweight fusion architectures enables emotion recognition systems that are both efficient and resilient, maintaining consistent performance without excessive computational cost [7].

2. Literature Survey

Research related to multimodal emotion recognition spans multiple technical directions, each addressing different aspects of affective modeling and cross-modal learning. Early approaches focused on combining text, audio, and visual features through feature-level or decision-level fusion, demonstrating that multimodal inputs significantly outperform unimodal baselines [1]. More recent studies employ deep learning architectures, particularly attention-based models, to learn joint representations that capture temporal and contextual dependencies across modalities [2]. Transformer-based fusion frameworks have shown strong performance by enabling direct interaction between heterogeneous modalities through self-attention and cross-attention mechanisms, improving emotion classification accuracy on benchmark datasets [3].

Another branch of research investigates architectural strategies for effective fusion and representation learning. Models such as cross-modal transformers and modality-invariant representation frameworks aim to align modality-specific embeddings while preserving complementary information [4]. Hierarchical and temporal attention mechanisms further enhance the modeling of long-range dependencies in multimodal sequences [5]. Despite their effectiveness, many of these architectures rely on deep transformer stacks and large parameter counts, leading to increased computational cost and limiting scalability [6]. Studies comparing fusion strategies highlight the trade-off between representational richness and efficiency, indicating the need for more compact fusion designs [7].

Earlier multimodal emotion recognition systems often assumed the consistent availability of all modalities during both training and inference. However, empirical analyses show that performance degrades substantially when one or more modalities are noisy or missing, a common scenario in real-world applications [8]. To address this issue, several works introduce robustness-enhancing techniques such as modality dropout, uncertainty modeling, and reliability estimation, which encourage models to adaptively adjust modality contributions [9]. While these methods improve resilience, they are frequently incorporated as auxiliary components rather than being central to the fusion process [10].

Beyond fusion and robustness, related research in adjacent areas provides valuable insights for multimodal emotion recognition. Studies on class imbalance and long-tail learning highlight the limitations of standard training objectives when emotion categories are unevenly distributed [11]. Data augmentation techniques, including audio perturbation, visual transformations, and text paraphrasing, have been explored to increase sample diversity and improve minority class performance [12]. Additionally, research on lightweight model

design and knowledge distillation demonstrates that compact architectures can retain much of the performance of larger models while significantly reducing computational overhead [13]. These findings align with the growing emphasis on efficiency and deployability in multimodal affective computing.
Despite these advances, the literature still lacks a unified framework that simultaneously addresses efficient fusion, robustness to modality degradation, and balanced recognition of low-resource emotion categories. Existing approaches tend to focus on individual aspects of the problem in isolation, resulting in fragmented solutions. This gap highlights the need for cohesive multimodal emotion recognition frameworks that integrate lightweight fusion, reliability-aware attention, and data-efficient learning strategies to support real-world deployment.

3. Existing System

Most existing multimodal emotion recognition systems are designed as tightly coupled pipelines that process text, audio, and visual inputs independently before combining them through a fusion module. Typically, modality-specific encoders extract features which are then aggregated using attention-based or concatenation-based fusion strategies, followed by a classifier to predict emotion labels. While such architectures demonstrate strong performance on benchmark datasets, they are often evaluated under controlled conditions where all modalities are assumed to be clean and consistently available [1]. This assumption limits their applicability in real-world scenarios, where one or more modalities may be noisy, incomplete, or misaligned. As a result, system performance degrades significantly when deployed outside laboratory settings [2].

Although automated deep learning models have replaced earlier handcrafted approaches, most current systems operate as isolated models trained for specific datasets or tasks. They lack mechanisms to adapt dynamically to varying input quality or distributional shifts. Some architectures incorporate auxiliary attention mechanisms or modality dropout to improve robustness; however, these additions are often treated as optional enhancements rather than integral components of the model design [3]. Furthermore, many systems rely on large transformer stacks and high-dimensional representations, leading to increased computational cost and inference latency. Without explicit consideration of efficiency, robustness, and data imbalance, existing multimodal emotion recognition systems remain constrained in their ability to balance accuracy with practical deployment requirements [4].

4. Proposed System

Recent research trends indicate a shift toward more efficient and resilient multimodal emotion recognition frameworks that address the limitations of existing systems. Contemporary approaches emphasize the integration of lightweight fusion architectures with adaptive attention mechanisms, enabling efficient cross-modal learning without excessive computational overhead [5]. In these systems, modality-specific encoders generate compact representations that are evaluated through reliability-aware components, which assess the quality of each modality before fusion. This process allows the model to down-weight noisy or unreliable inputs while preserving informative signals [6].

To improve generalization across diverse emotion categories, especially underrepresented ones, targeted data augmentation strategies have been explored. These methods selectively enhance minority emotion classes through modality-specific perturbations and balanced sampling techniques, improving training stability and reducing prediction bias [7]. In addition, recent studies advocate incorporating augmentation-aware objectives directly into the training pipeline rather than treating augmentation as a preprocessing step [8]. Such integration enables models to learn more robust representations that generalize better across varied conditions.
Architecturally, the reviewed systems favor modular designs consisting of compact

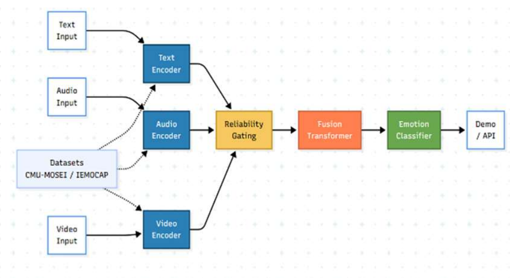
Figure 1 Architecture of the proposed platform encoders, reliability-aware fusion modules, and lightweight classifiers. These components collectively support scalability, interpretability, and robustness while maintaining manageable computational complexity [9]. By combining efficiency-oriented design with adaptive fusion and data-centric learning strategies, the proposed direction addresses key shortcomings of existing multimodal emotion recognition systems and aligns more closely with the demands of real-world human–AI interaction applications [10].

5. System Architecture

The reviewed architectures for multimodal emotion recognition are organized into multiple coordinated components that collectively support efficient cross-modal learning and robust emotion inference. Typically, raw inputs from text, audio, and visual modalities are first processed by modality-specific encoders that extract compact semantic, acoustic, and facial representations [19]. These representations are then evaluated by reliability-aware modules that estimate the quality of each modality, allowing the system to account for noise, missing data, or modality degradation before fusion [20]. By explicitly modeling modality reliability, the architecture avoids over-reliance on unreliable inputs while preserving informative signals.

As illustrated in Figure 1, the fusion module integrates the reliability-weighted representations using lightweight transformer-based mechanisms. This fusion process enables contextual interaction across modalities while maintaining computational efficiency [21]. Temporal modeling and attention-based weighting further ensure that transient noise or isolated artifacts do not disproportionately influence the final emotion representation, leading to more stable and consistent predictions across varied input conditions.

To facilitate practical deployment, many architectures expose the emotion inference pipeline through a modular interface that supports real-time prediction and downstream integration. Such interfaces allow applications with limited computational resources to leverage multimodal emotion analysis without maintaining complex internal models [22]. Through this layered design, the system supports robustness, scalability, and adaptability, enabling reliable emotion recognition across diverse real-world scenarios while maintaining manageable computational overhead.



6. Conclusion

The analysis presented in this review highlights the limitations of existing multimodal emotion recognition systems, particularly their dependence on computationally intensive architectures and assumptions of ideal input conditions. These constraints reduce robustness under noisy or missing modalities and hinder balanced recognition of underrepresented emotion categories. Although transformer-based fusion models have advanced the state of the art, their practical deployment remains challenging due to efficiency and generalization concerns.

By synthesizing recent research trends, this study emphasizes the importance of lightweight fusion architectures, reliability-aware attention mechanisms, and targeted data augmentation strategies. Systems designed along these principles are better positioned to achieve consistent performance across diverse environments while remaining computationally feasible. As multimodal learning techniques continue to evolve, frameworks that balance accuracy, robustness, and efficiency are likely to play a central role in advancing real-world emotion-aware human-AI interaction systems.

7. References

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